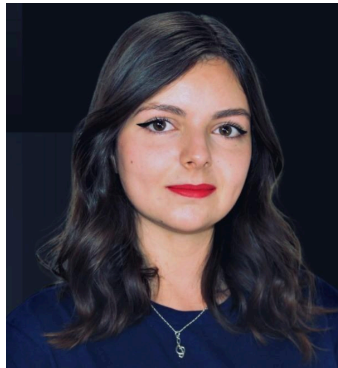




MÉLODIE BOULY

DATA SCIENTIST

4 YEARS OF EXPERIENCE

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27 ans

323 Rue de l'aiguelongue 34090 Montpellier

French citizen – willing to relocate in Switzerland

<https://www.linkedin.com/in/m%C3%A9lodie-bouly-03b8641a3/>

OBJECTIVE

Data Scientist with 4 years of experience in machine and deep learning applied to medical imaging. Specialized in developing processing pipelines and segmentation and classification models for complex medical data. Strong analytical mindset, scientific rigor, and proven ability to collaborate within multidisciplinary teams in the fields of digital health and research. I am looking to settle permanently in Switzerland.

KEY TECHNICAL SKILLS

- **Machine Learning** (sklearn, xgboost, Flaml, Automl)
- **Deep Learning** (Pytorch, Tensorflow, MONAI, Monailabel)
- **Data manipulation, analysis and visualization** (numpy, pandas, matplotlib, seaborn, plotly)
- **Big data** solution usage (aws)
- Technical english

SOFT SKILLS

- Team spirit
- Rigor and attention to detail
- Strong willingness to learn and contribute
- Open-mindedness, empathy, and strong adaptability

PROFESSIONNAL EXPERIENCE

SIM&CURE
Junior Data Science Engineer
 Montpellier (France)

Since Oct 2022
 3 years

Medical sector

12 mois

Context: Data Scientist in a team of 4 within the Research & Development department of a 60-person startup.

Tasks:

- Independently developed a machine learning model for predicting aneurysm rupture status, based on clinical and morphological features.
- Collaboratively developed a two-step deep learning pipeline for detecting intracranial aneurysms on MRA and CTA scans (data cleaning, statistical analysis, coding for network training, model evaluation, hyperparameter optimization).
- Independently developed an annotation initialization plugin for the 3DRA modality as part of a partnership with a consortium of European hospitals, along with a transformer-based model for the segmentation of cerebral arteries and intracranial aneurysms on 3DRA, integrated into this plugin.
- Conducted a solo project for analyzing and visualizing usage data from the company's flagship software via an API, presenting insights to executives and relevant divisions to support decision-making.
- Supervised two internships (one Master's and one Bachelor's student).

Technical environment :

Python / AWS / Pytorch / sklearn / pandas / numpy / seaborn / pyvista / ci/cd / gitlab

SIM&CURE Data Scientist en apprentissage Montpellier (France)	Sept 2020 - Oct 2022 : 2 ans
Medical Sector 12 mois Context: Data Scientist apprentice during a Master's work-study program, part of a 3-person team within the Research & Development department of a 30-person startup. Tasks : • Performed data engineering on complex medical datasets. • Developed and evaluated a machine learning model deployed in the company's software to automate an arterial reconstruction task. • Collected data in hospitals and collaborated with physicians. • Conducted data visualization and analysis. Technical environment : Python / AWS / Pytorch / sklearn / pandas / numpy / seaborn / pyvista	

EDUCATION

Master's Degree in Data Science (Work-Study Program)

Université de Montpellier (France)

2020-2022

Bachelor's Degree in Computer Science

Université de Montpellier (France)

2017-2020

LANGUAGES

- French : fluent
- English : Professionnal
- German : B1

REFERENCES

- Christophe Chnafa, Chief Technical Officer at Sim&Cure : +33 6 65 04 85 69
- Yoann Pitarch, Data Science Manager at Sim&Cure : +33 6 15 47 26 36
- Lorenzo Posti, Expert Data Scientist at Sim&Cure : +33 6 67 68 64 41

INTERESTS

- Dance (Salsa, bachata, rock)
- 4 musical instruments
- Reading
- Swiss !